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Macroeconomic analysis of the employment impacts of future EU climate policies

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This article gives a detailed account of part of the modelling that was carried out for the assessment of the EU's proposed energy and climate targets for 2030. Using the macro-econometric simulation model, E3ME, and drawing on results from the PRIMES energy systems model, it shows that a 40% reduction in GHG emissions (compared to 1990 levels) could lead to an increase in employment of up to 0.7 million jobs in Europe. Furthermore, if the same GHG reduction target was combined with targets for renewables and energy efficiency, the net increase in jobs could be as high as 1.2 million. Both results are in contrast to the standard findings from computable general equilibrium (CGE) models, reflecting the different underlying assumptions (e.g. labour supply) to the modelling approach. Additional sensitivity testing shows that the ways in which the energy efficiency and renewable measures are funded are important factors in determining overall economic impact.

Policy relevance

In recent years there has been much debate as to whether the European Union should have a single GHG reduction target or a set of targets that also cover renewables and energy efficiency. This paper elaborates on part of the modelling that was carried out for the official assessment of the European Union's proposed energy and climate targets for 2030. Using an empirical, model-based approach, it compares a scenario where there is a single 40% GHG reduction target to a scenario that also includes a 30% renewables target and stricter energy efficiency standards. The model results show that the large investment stimulus needed to meet the combined targets leads to higher levels of GDP and employment. This suggests that there could be medium-term economic and social benefits to including all three targets in the future energy and climate package.

Keywords: climate policy; employment; energy efficiency; macroeconomic modelling; renewables

1. Introduction and background

This article assesses the consequences for employment of two possible ways of meeting the target of reducing GHG emissions in the EU by 40–45% by 2030, compared to 1990 levels. The target was proposed by the European Commission (EC) in its policy framework for climate and energy in the period 2020–2030 (EC, 2014). EU leaders have agreed to take a final decision on it by October 2014. The policy framework and its targets are part of the exploration of ways of reducing GHG emissions initiated as a follow-up to the 2050 roadmap towards a low carbon economy (EC, 2011, 2013).

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The current legal framework for climate and energy policy to 2020 contains three elements:

- A 20% reduction in the EU's GHG emissions by 2020 compared to 1990 levels
- A target (agreed in 2009) that 20% of gross inland energy consumption should be met by renewables by 2020
- A non-binding, aspirational target to improve energy efficiency by 20% in 2020, compared to the energy consumption expected in 2020 under the baseline projection made in 2007

Beyond 2020, the EU's climate policy is shaped by the 2 °C target. The estimates made by the Intergovernmental Panel on Climate Change (IPCC) suggest that developed countries would need to reduce GHG emissions by 80–95% by 2050 to meet this target. In the 2050 roadmap (EC, 2011) the EC argued that, if the EU is to meet the 80–95% target in a cost-effective way by 2050, it will need to reduce its GHG emissions by 40% in 2030, compared to 1990 levels. Moreover, the argument continued, to postpone taking action would be more costly, because more expensive reductions would be needed in the final decades before 2050 to compensate for the lack of action in earlier periods. That is the rationale that underlies the suggested target of a 40% reduction by 2030 proposed in the roadmap. This article assesses the consequences for the use of that target.

We consider two decarbonization policy scenarios that have been developed as background analysis for the 2030 framework. Each scenario defines policy measures at the EU28 level over the period 2010–2030 and is assessed using the E3ME macroeconomic model (Cambridge Econometrics, 2014), on the basis of previous results from the PRIMES energy system model (E3MLab of ICCS/NTUA, 2010). The two decarbonization scenarios are compared to a reference case that reflects likely trends in energy investment, by scale and mix, in the absence of the additional measures. In both decarbonization scenarios there is a reduction in CO₂ emissions equivalent to a reduction in GHG emissions of 40% from 1990 levels. The second decarbonization scenario also contains a more ambitious energy-efficiency policy and a target that 30% of gross final energy consumption should be met by renewables in 2030.

The following section gives a brief overview of previous assessments of the employment effects of climate policy. Section 3 of the article explains the methodology employed and Section 4 presents and discusses the results. Section 5 concludes.

2. Previous analyses of the employment effects of climate policy

The IPCC's Fourth Assessment report (Barker et al., 2007, p. 767) states that investment in GHG mitigation could have a greater impact on employment than investments in conventional technologies. Previous analysis has shown that energy efficiency can induce a positive effect on total employment, even after taking into account all direct and indirect factors, and a 20% saving in energy consumption in the EU could create up to one million new jobs in Europe. Recycling carbon tax revenues via social security contributions could increase employment by nearly 1% by 2010 in France and Germany, but much less in the US and Japan. A study of climate policies for California also reports small increases in employment for a package of measures focusing on the tightening of regulations affecting emissions.

More recently, Fankhauser, Sehleier, and Stern (2008) found that the impact of climate policy on jobs is complex. In the short term, the net effect could be job creation, as low-carbon technologies

tend to be more labour-intensive. In the medium term, the effect is felt economy-wide, but is more difficult to gauge, particularly if climate policy is unilateral and trade effects are taken into account. However, the biggest effect is expected to be long term, when climate policy will trigger the widespread 'creative destruction' often associated with innovation, job creation, and growth.

Delbeke, Klaassen, van Ierland, and Zapfel (2010) have shown that the EU's climate policy for 2020 could increase employment in the EU in 2020 by 0.05% or reduce it by 0.09%, depending on how targets are distributed among Member States and whether access to the Clean Development Mechanism (CDM) is allowed. Klaassen, Nill, van Ierland, Saveyn, and Vergote (2012) found that a 30% GHG reduction in the EU in 2020, with limited flexibility and the Copenhagen Accord, could result in small positive employment impacts if auctioning and carbon taxation are introduced that recycle the revenue to lower labour costs. Böhringer, Keller, & Van der Werf (2013) found that job creation from renewable energy sources in Germany may be limited and crucially depends on the level of the subsidy and the financing mechanism.

Neuwahl, Löschel, Mongellin, and Delgado (2008) conclude that the EU's biofuel targets of 10–15% could be achieved without adverse net employment effects. Although papers do exist that examine the welfare losses of climate policies in combination with policies that promote renewable energy (e.g. Böhringer, Rutherford, Balistreri, & Weyant, 2012), no economy-wide studies appear to exist that examine the employment impacts of realistic EU-wide climate policies combined with renewable and energy-efficiency policies. This article attempts to fill the gap.

3. Methodology: linking a macroeconomic model with an energy systems model

3.1. Overview of the E3ME macroeconomic model

E3ME is a macro-econometric simulation model that combines the economy and labour markets with Europe's energy systems and related GHG emissions. The model has been applied in the past for economic analysis of European energy and climate policy, including assessments of green jobs (Cambridge Econometrics, GHK and IER, 2011), the EU Energy Efficiency Directive (Cambridge Econometrics and ECN contribution to European Commission, 2011) and the EU's Energy Roadmap (Cambridge Econometrics, E3MLab NTUA, E&Y, Exergica and IER, 2014). The following characteristics are important for this analysis:

- It covers each of the European Member States at a national level
- It has a detailed sectoral specification
- Its econometric specification is based on a strong empirical grounding
- It has a detailed treatment of labour market effects
- It incorporates physical flows of energy in its structure

Further information, including the full model manual (Cambridge Econometrics, 2014) and publication list, is available online at www.e3me.com. We provide a brief outline in the following section, and then discuss three key issues relevant to this article (capital markets, household consumption, and labour markets).

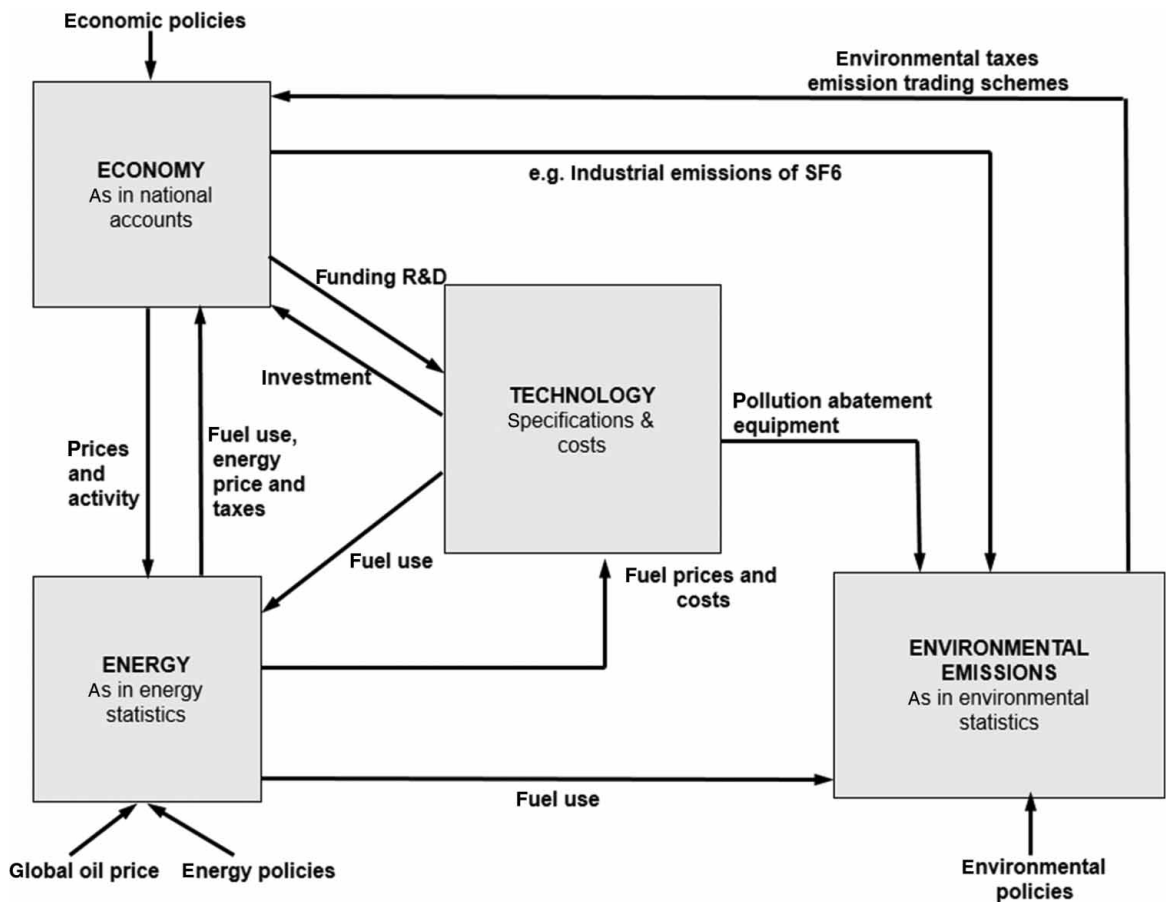


Figure 1 E3ME modules
Source: E3ME.

Figure 1 shows the main modules in E3ME. The economic structure of E3ME is based on the system of national accounts, as defined by ESA95 (EC, Eurostat, 1996). Short-term multiplier effects occur through the various interdependencies and feedback loops that are present in the model structure. There are 33 sets of econometrically estimated equations, including the components of gross domestic product (GDP; consumption, investment, and international trade), the labour market, prices, energy demand, and materials demand. Each equation set is disaggregated by country and by sector.

The economy and energy demand are closely linked; economic activity creates the demand for energy, but energy consumption also affects the economy through output in the energy extraction and distribution sectors (e.g. electricity, oil and gas). Most environmental emissions are caused by fuel combustion (modelled as fixed coefficients), but there are also direct economy-emission linkages through process emissions.

Technology, which is endogenous in E3ME, can affect many of these relationships. For example, the use of energy-efficient vehicles allows an increase in economic production without an increase in energy consumption and emissions. Some particular technologies like carbon capture and storage (CCS) or renewables allow energy consumption to increase without increasing emissions.

The main dimensions of the E3ME model are as follows:

- 53 countries (limited in scope to the EU28 Member States for this study)
- 69 economic sectors, defined at NACE 2-digit level and linked by input–output relationships
- 42 categories of household expenditure
- 13 types of households, including income quintiles and socio-economic groups such as the unemployed, inactive, and retired, plus an urban/rural split
- 22 different users of 12 different fuel types
- Coverage of the six Kyoto GHGs, and other emissions where available

The E3ME model is macro-econometric in design. Although it shares many of the characteristics of a computable general equilibrium (CGE) model, there are some important differences in approach. In general, CGE models assume full price adjustment, optimization, and equilibrium in all markets. Therefore, the results tend to be determined by the supply-side conditions such as resource availability and labour supply. In other words, it follows from the modelling assumptions that the models yield no demand-side effects of policy change on consumption or private investment and consequently there are no positive effects on economic performance in the model results.

In contrast, the E3ME model is post-Keynesian in nature. It is a macro-econometric model based on the theory of effective demand in which there may be available and unused resources in the economy. A boost to demand, for example through higher levels of investment, could lead to higher levels of economic activity and employment, including supply chain and multiplier effects. The possibility that additional regulation could stimulate the economy is in contrast to the more standard findings from CGE models (see Edenhofer et al., 2010, for a discussion), but this is also linked to the assumption the modellers make on (rigid) labour supply.

3.2. Key modelling issues

Three specific cases of these non-optimization assumptions are particularly relevant for the scenarios described in this article.

The first is the treatment of capital markets. Although E3ME respects the accounting identity that savings and investment match, in contrast to CGE models, higher investment in energy technologies does not necessarily lead to a ‘crowding out’ of investment elsewhere. The model allows, for example, the creation of money by the banking sector and/or the sale of financial assets to finance fixed investment in new assets. Investment in new assets stimulates real economic activity and therefore results in GDP and employment growth, which could in turn provide the financing for further investment. In summary, if policy provides the stimulus for additional investment in new assets, this could lead to higher rates of overall activity.¹

The second is the treatment of household consumption and wealth. E3ME includes an assumption that, in the long run, changes in income and expenditure will match. So, when income tax rates are

increased to pay for energy investments, there will be a corresponding equivalent reduction in household consumption. However, there is no similar constraint on wealth. In the scenario results, net levels of household wealth may change due to the following:

- Short-run fluctuations in savings rates; as noted above, in the long run, incomes and wealth are assumed to move together, but a household's immediate response to a change in income may be to run down savings to maintain previous levels of expenditure
- Direct impacts through the lump-sum charges imposed to pay for energy sector investments (described in further detail in Section 3.5)

To sum up, in common with most other economic models, the focus on E3ME is very much on income flows, rather than the stock of wealth. However, the model structure, and the particular assumptions adopted in this analysis, allow for the possibility of transfers to and from wealth stocks to retain or boost current levels of real economic activity. The focus on wealth in this study partly reflects the parallels with the stock of carbon in the atmosphere that has built up over time, but also recent interest in wealth taxes for reasons of social equality.²

The third issue, labour markets, is of particular relevance to this article. Under the basic market clearing assumptions of a standard CGE model, wage rates will fully adjust so that labour supply and demand are matched and there is no unemployment. In E3ME there are econometric equations estimated for labour supply (by age group and gender), employment demand (by sector), and wage rates (also by sector).

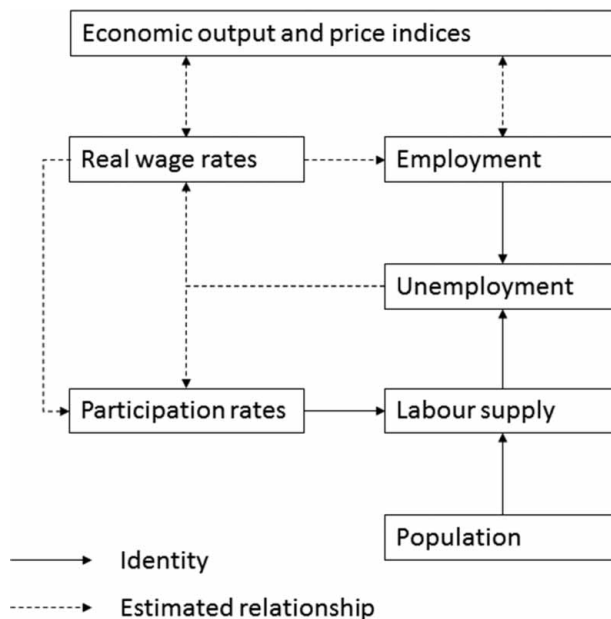


Figure 2 Outline of the E3ME equations

Figure 2 summarizes how the different equations fit together. Movements in wage rates are key to determining overall employment outcomes; if higher economic demand leads to higher wage rates then employment levels may not change by much, but if wages do not adjust then the impact on employment will be higher. It should be noted that in both cases there may be an increase in labour supply as both higher wages and lower unemployment rates can lead to higher participation rates and draw more workers into the labour market.

3.3. The reference case

The scenario inputs for this analysis are taken from results from the PRIMES energy systems model (NTUA, 2010). There is one reference case and two decarbonization scenarios.

Our modelling baseline is the new PRIMES Reference Scenario 2012–2013. This includes all current and planned policies to 2020 that are necessary to achieve the EU targets for renewable energy sources (RES) and GHG emissions (Capros et al., 2013). In the reference case it is assumed that beyond 2020 there are no GHG emissions reduction targets for EU countries and any improvements in GHG intensity are due to the policies already adopted. It is regarded as a ‘business as usual’ (BAU) case, in that it represents expected outcomes if no further policy is implemented. Because this BAU is the standard modelling baseline for use in EC studies, it provides a degree of consistency across research projects. Figure 3 depicts emissions levels for the reference case.

In order to ensure consistency in approach, we matched the E3ME and PRIMES reference cases as closely as possible. The E3ME reference case was set to match the PRIMES outputs for economic projections, energy prices, emissions trading scheme (EU ETS) prices, projections of energy demand by sector

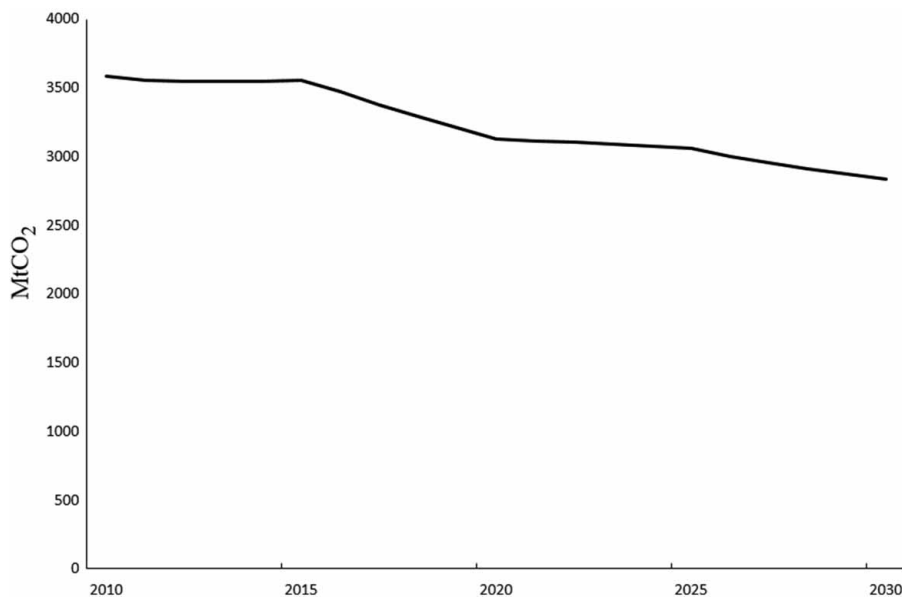


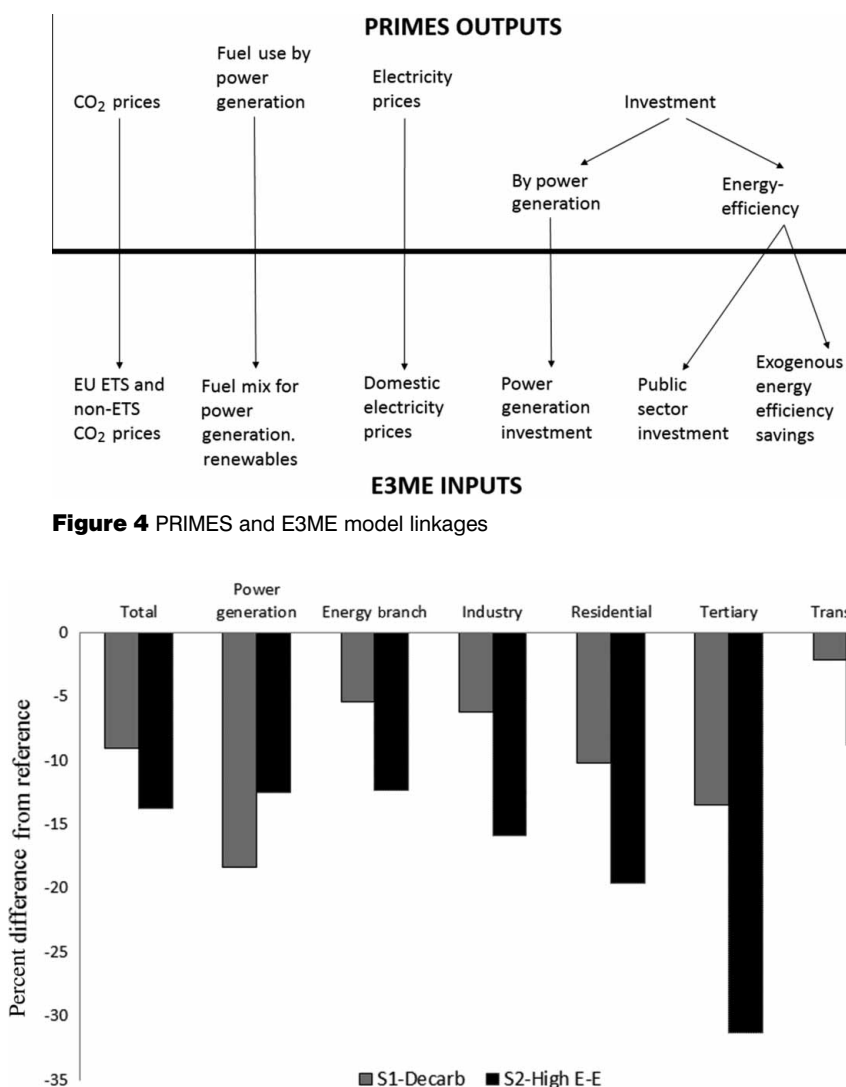
Figure 3 EU28 CO₂ emissions, reference case

Source: PRIMES model.

and by fuel, sectoral CO₂ emissions, and electricity capacity and generation. These were processed to be consistent with the E3ME sectoral classifications and interpolated to provide annual time series.

3.4. The decarbonization scenarios

For the decarbonization scenarios, E3ME incorporates the energy system outputs from the PRIMES model, and uses these to make projections of economic activity and employment. The outputs from



the PRIMES model are described in EC (2014), but, for the purposes of this article, are otherwise taken as given. Figure 4 summarizes the model linkages, which are discussed in more detail in the following.

In both decarbonization scenarios there is a reduction in CO₂ emissions equivalent to a reduction in GHG emissions of 40% in 2030 from 1990 levels (see Figure 5 for a sectoral breakdown). This is achieved through a combination of carbon prices, developments in the electricity sector, and investment in energy efficiency. The second decarbonization scenario (S2) contains a more ambitious energy-efficiency policy and a target that 30% of gross final energy consumption should be met by renewables in 2030.

The main inputs to E3ME for the scenarios are as follows:

1. Changes to carbon price policy (i.e. introduction of a carbon tax for non-ETS sectors and, in the first decarbonization scenario, higher ETS allowance prices)
2. A shift in the power generation mix away from conventional fossil fuels towards renewable sources
3. Energy savings resulting from additional investment in energy-efficiency improvement in the EU28

In both the reference case and the two decarbonization scenarios, the revenues raised from power generation (through the ETS) and from non-ETS sectors (through a carbon tax) are recycled (see discussion in Section 4.5). Differences between the scenarios with respect to these measures are therefore revenue-neutral, with the consequence that the economic results presented in Section 4 reflect a shift in resources rather than an increase or decrease in the overall level of taxation. In line with existing legislation, allowances to the industrial ETS sectors are allocated for free and do not generate income.

Although E3ME could be used to assess energy consumption (including estimating the carbon prices needed to meet the emission reduction targets), it does not have the same level of detail as the PRIMES energy model. Several of the scenario inputs are therefore taken directly from the PRIMES outputs:

- The EU ETS carbon price and the carbon tax rates on non-ETS sectors (from 2025 onwards)
- The change in electricity prices (which pays for the renewables investment)
- The power generation mix and the level of CCS

This means that we implicitly adopt some of the modelling assumptions used by PRIMES, such as the translation from emission reduction targets to carbon prices and also the development path of the carbon prices over the projection period. It is also noted that, under current circumstances, a carbon tax on the non-ETS sectors seems unlikely to be feasible politically. In the PRIMES analysis, the tax (more accurately the shadow price of carbon) is a modelling assumption made so that the emission reduction targets are met (with reductions from all sectors) rather than a statement about future policy.

The direct costs of investment in energy efficiency are also based on PRIMES results. The energy efficiency data cover the EU28 for both the residential and tertiary sectors. The monetary values are converted into units of energy savings using a coefficient derived from the IEA's *World Energy Outlook* (IEA, 2012).³ The values for the residential sector were translated into energy savings made by households, while the values for the tertiary sector were disaggregated into energy savings made by agriculture and services. The energy savings were further disaggregated based on the PRIMES results provided, and were also shared between energy carriers in proportion to energy consumption (in order to obtain a more detailed fuel split). Finally, the energy saving per unit of investment was calibrated to make the

Table 1 Carbon prices

	2015	2020	2025	2030
<i>Reference case</i>				
ETS sectors	5	10	14	35
<i>S1 – decarbonization</i>				
ETS sectors	5	12	18	40
Non-ETS sectors	0	0	18	40
<i>S2 – high energy efficiency</i>				
ETS sectors	5	10	10	11
Non-ETS sectors	0	0	10	11

Source: PRIMES.

Note: Prices in €₂₀₁₀/tCO₂.

outcome for final energy use in the scenario broadly match the reduction from the PRIMES decarbonization scenarios in percentage terms.

It is assumed that the energy efficiency investment is funded directly via government subsidy and is additional to any efficiency improvements made in the baseline. This investment therefore affects the public budget and either requires the introduction of a revenue-raising instrument or reduces the revenues available for recycling from auctioned ETS allowances or carbon tax receipts (see Section 3.5).

Table 1 shows the carbon prices used in the reference case and the two decarbonization scenarios. In the first decarbonization scenario the carbon price increases gradually over time to meet the 40% GHG

Table 2 Changes in net installed power capacity, 2010–2030 (GW)^a

Energy technology	Reference	S1 – decarbonization	S1 difference from reference	S2 – high energy efficiency	S2 difference from reference
Wind	221	247	25.9	267	46.0
Solar photovoltaic	120	126	5.5	129	8.5
Oil	– 30	– 32	– 1.7	– 33	– 3.2
Gas	56	37	– 19.1	25	– 31.4
Coal	– 72	– 72	0.4	– 75	– 3.3
Additional capacity	300	315	15.2	326	26.5

Source: PRIMES.

Note: ^aThis includes power generation capacity derived from new power plants, premature replacement, and retrofitting.

Table 3 Additional investment in energy-efficiency (in €₂₀₁₀ billion, five-year totals)

	2015–2020	2020–2025	2025–2030
<i>S1 – decarbonization</i>			
Residential	3	122	148
Tertiary	4	101	118
Total	7	223	266
<i>S2 – high energy efficiency</i>			
Residential	8	95	219
Tertiary	1	99	263
Total	9	194	482

Source: PRIMES.

reduction in 2030. The difference between the carbon price in this scenario and in the reference case is not very large at any time. In addition, the reference case is already expected to reduce the EU's GHG emissions by some 32% in 2030. In comparison to the present level, however, the carbon price of €40/tCO₂e in 2030 is eight times higher. The carbon prices in the second decarbonization scenario are lower than in the reference case, as the ambitious energy-efficiency policies and the target that 30% of gross final consumption should be met by renewables lead to lower fuel demand and CO₂ emissions and, thus, a lower carbon price.

Table 2 summarizes the PRIMES results for the power sector. It is important to note that the price of electricity is affected by changes in carbon prices as well as by the switch to renewables in the power generation sector and any energy-efficiency measures that may be implemented. As noted above, this effect is also taken from the PRIMES results, and added exogenously in E3ME. In the baseline, electricity prices increase by 30.8% in real terms over the period 2010–2030; in S1 the figure is 33.3% and in S2 it is 32.2%.

Table 3 summarizes the additional investments in energy efficiency improvements as five-year totals up to 2030 in comparison to the reference projection. Clearly, the additional investments in energy efficiency improvements compared to the reference case are significant, particularly between 2025 and 2030.

3.5. Revenue recycling

Equation (1) shows the factors from which the direct impact on government balances is calculated in the decarbonization scenarios:

$$\begin{aligned}
 \text{Change in balance} = & \text{revenues from auctioned ETS allowances} \\
 & + \text{revenues from carbon taxes (after 2025)} \\
 & - \text{public investment in energy efficiency}
 \end{aligned} \tag{1}$$

Because these scenarios are designed to be revenue-neutral, a further adjustment must be made to ensure that the (direct) change in government balance is zero. Our results show that the chosen method of adjustment is an important determinant of final outcomes.

In our main results we assume that, when the change in government balance is positive (i.e. the investments are less than the revenues raised), this difference is given to households each year in the form of a lump sum payment. This approach is commonly used in modelling exercises because the payments, being equal to all households, do not have incentive effects on consumption choices and therefore the approach is less distortive than alternatives. In the two decarbonization scenarios discussed in this article, the lump sum payments are treated as a change in wealth rather than a change in income. This assumption is important, as all income is spent (most of it immediately), but wealth is allowed to accumulate and only a small proportion is spent immediately. For this modelling exercise we have assumed that only 2.5% of the amount received by consumers is spent immediately, a figure that is broadly in line with the ratios used in the existing literature.⁴

When the balance is negative (i.e. the investments are greater than the raised revenues), the shortfall is financed through lump sum payments made by households. This is similar in nature to a tax on wealth.

We also assessed an alternative method of revenue recycling, using changes in labour tax rates instead of lump sum payments. These results are shown in Section 4.5.

4. Employment impacts of the 2030 decarbonization scenarios

4.1. Factors for which the E3ME modelling is expected to show impacts

Figure 6 provides an overview of the economic factors for which impacts are expected to be shown in the E3ME modelling under the two decarbonization scenarios. The main macroeconomic indicators that are affected are as follows:

- Investment: expected to increase as a result of (1) the required shares of renewables in energy generation and (2) levels of energy efficiency
- Household expenditure: will change as a result of changes in product prices (including electricity) and revenue recycling
- Net trade: will change as a result of competitiveness effects and changes in fossil fuel imports

The sum of these macroeconomic changes will produce the GDP impact and this, in turn, will lead to changes in employment.

Inflation rates would also be expected to increase in the first decarbonization scenario, reflecting the higher electricity and product prices. In the second decarbonization scenario, however, they could either increase or decrease because electricity prices will be higher but ETS carbon prices will be lower.

Barker, Ekins, Junankar, Pollitt, and Summerton (2009a) and Barker, Anger, Chewpreecha, and Pollitt (2012) explain further how the impacts are determined within the modelling framework.

4.2. Macroeconomic performance

E3ME produces results for 69 economic sectors. These include all the direct and indirect effects (of which the main ones were described in Section 4.1) of the scenario inputs (displayed in Figure 6).

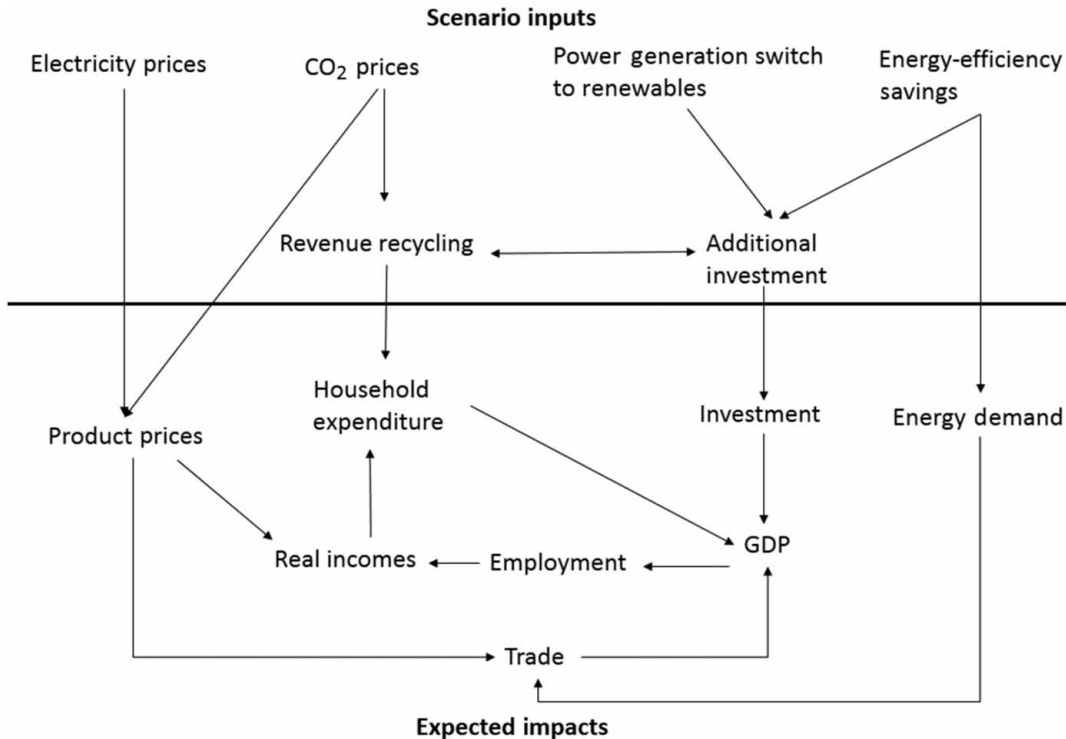


Figure 6 Impacts of scenario inputs on results of E3ME modelling

Table 4 shows the main results by economic indicator for 2030. The policies in both decarbonization scenarios result in a small increase in overall levels of economic activity (measured by GDP) and household expenditure. Investments on a considerable scale are required to meet the 40% emissions reduction target and the energy-efficiency (and, to a lesser extent, renewable)

Table 4 EU28 summary macro indicators, 2030

Indicator	Reference case (€billion 2005 prices)	S1 – decarbonization (% increase over reference case)	S2 – high energy efficiency (% increase over reference case)
GDP	15,778	0.6	1.1
Consumer expenditure	8,885	0.1	0.5
Investment	3,869	2.1	3.5
Exports (extra-EU)	2,756	0.0	0.1
Imports (extra-EU)	2,831	0.2	0.5
Price index (2005 = 1)	1.5	0.5	– 0.1

Source: E3ME.

targets. In the first decarbonization scenario an additional €266 billion is invested in energy efficiency over the period 2025–2030, while in the second decarbonization scenario the additional investment in energy efficiency amounts to €482 billion over the same period (see Table 3 for a breakdown of these totals). These sums are large enough to have a noticeable impact at the macro level.⁵

In the second decarbonization scenario there is further investment due to the higher energy efficiency requirements. There is also a benefit to Europe's economy as the general price level is very slightly lower compared to the reference case, a direct result of the lower carbon prices arising from the higher energy-efficiency savings. This boosts real incomes and consumer expenditure.

4.3. Net aggregate employment impacts

For most sectors, employment levels in E3ME are modelled as a function of economic output, relative labour costs, technology, and working hours (see Appendix or Cambridge Econometrics, 2014, for a formal definition of the equation). There is little change to the state of technology or level of working hours in the scenarios, so the employment results largely reflect trends in economic output and wage rates.

Until 2020 there is not much difference in employment in either decarbonization scenario compared to the reference case, because the new policies in the decarbonization scenarios are introduced after 2020. In all scenarios, employment is projected to increase by around 3.5% over the period 2012–2020, mainly due to recovery from recession. This 3.5% increase includes all employment effects resulting from economic projections and all current and planned climate and energy policies to 2020.

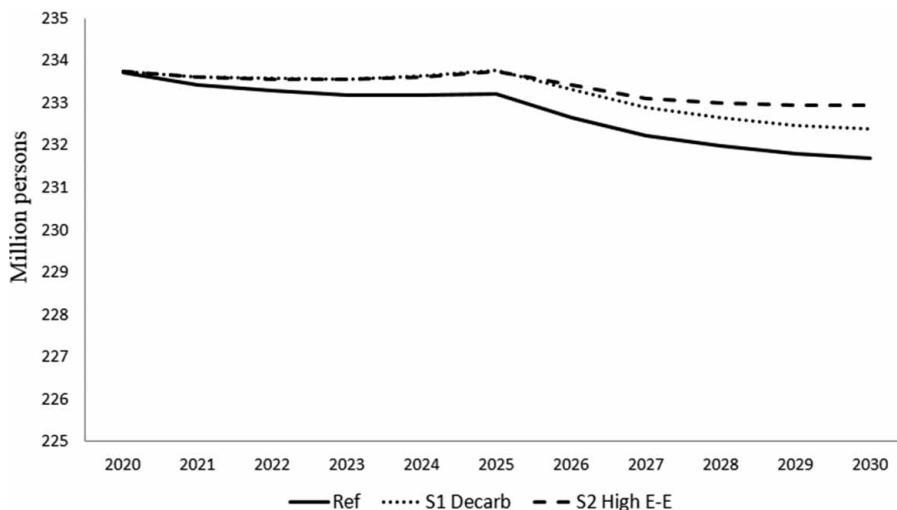


Figure 7 EU28 total employment in the reference case and decarbonization scenarios

Source: E3ME.

Note: Decarb, decarbonisation scenario, High E-E, high energy efficiency scenario.

Table 5 EU28 employment (in 1000s of persons, headcount)

	2010	2015	2020	2025	2030
Reference case	226,149	231,910	233,710	233,222	231,701
S1 – decarbonization	226,149	231,911	233,737	233,785	232,379
Difference from reference case	0	1	27	563	678
% Difference	0.0	0.0	0.0	0.2	0.3
S2 – high energy efficiency	226,149	231,912	233,754	233,757	232,947
Difference from reference case	0	2	44	535	1,246
% Difference	0.0	0.0	0.0	0.2	0.5

Source: E3ME.

After 2020, the decrease in working age populations across Europe is expected to lead to a slow decline in employment in the reference case, and after 2025 the pace of decline is expected to quicken somewhat (see [Figure 7](#)). In both decarbonization scenarios, on the other hand, the net impact on employment is positive, as a result of the higher level of economic activity (see [Figure 7](#) and [Table 5](#)). At the end of the period 2020–2030, there are 678,000 more jobs in the first decarbonisation scenario than in the reference case (a positive difference of 0.3%), while in the second (high energy efficiency) decarbonization scenario there are 1,246,000 more jobs than in the reference case (a positive difference of 0.5%). Real wage rates are almost unchanged in the first scenario but increase by 0.2% in the second scenario.

The higher employment in both decarbonization scenarios is not sufficient to prevent an overall decline in Europe's employment levels over 2025–2030. Nevertheless, in both scenarios, the decline is delayed until after 2025 and slowed somewhat thereafter. The differences between the two scenarios in effects on employment are largely driven by the energy efficiency investment and the carbon tax that is introduced in 2025. The second decarbonization scenario, because of its higher investment in energy efficiency, has more positive effects on employment, in line with the outcomes for GDP.

The predicted employment outcomes can be explained mainly by the projections of the overall level of economic activity (i.e. output). However, there are some other contributing factors, albeit with smaller impacts overall. These factors include the responses of wage demands to changes in prices, substitution rates between energy and labour, and other local labour market conditions.

4.4. Employment impacts by sector

The aggregate employment effects mask the variation in patterns of structural change at the sectoral level. The overall pattern of structural shifts in both decarbonization scenarios is characterized by an increase in employment in the investment-related sectors, and decreases in energy and some energy-intensive sectors (see [Table 6](#)).⁶ This section discusses these variations for the two scenarios.

The main points are that investments in capacity for energy generation from renewables and in energy-efficient equipment and technologies create jobs in construction, mechanical and electrical

Table 6 Employment impact of EU climate package (S1) and high energy efficiency scenario (S2) in 2030 by sector (in 1000s of persons)

	Reference case	Decarbonization scenario S1	Difference from reference case (S1)	High energy efficiency scenario S2	Difference from reference case (S2)
Agriculture	9,391	9,402	11	9,407	16
Extraction industries	500	479	– 21	498	– 2
<i>Of which: coal extraction</i>	139	117	– 22	136	– 3
<i>Of which: oil and gas extraction</i>	110	110	0	110	0
Basic manufacturing	14,839	14,913	74	14,944	105
Engineering and transport equipment	15,277	15,367	90	15,429	152
Utilities	2,280	2,301	21	2,308	28
Construction	16,599	16,708	109	16,890	291
Distribution and retail	35,314	35,348	34	35,452	138
Transport	9,411	9,455	44	9,471	60
Communications	20,307	20,384	77	20,440	133
Business services	41,048	41,225	177	41,293	245
Public services	66,735	66,797	62	66,814	79
Total	231,701	232,379	678	232,947	1,246

Source: E3ME.

engineering, and their supply chains (see Table 6). The construction sector and its suppliers, because they benefit directly from additional investment in energy efficiency and power generation (e.g. building retrofits), are expected to have 109,000 more jobs in 2030 than in the reference case. Mechanical and electrical engineering also benefit directly from the switch to renewable technologies as well as from the drive to increase energy efficiency. There are expected to be around 90,000 more jobs in engineering and transport equipment in 2030 than in the reference case.

Employment in utilities is also expected to increase (although remain small), because renewable technologies for energy generation can be more labour-intensive than conventional methods of energy generation (Cambridge Econometrics, E3M-Lab NTUA, E&Y, Exergia and IER, 2014; Wei, Patadia, & Kammen, 2010). However, employment in the energy extraction industries and some parts of the utilities sector (e.g. gas supply) is sensitive to the net impact of offsetting effects. In the decarbonization scenario, these are produced either by higher carbon prices or higher energy prices, or by both. The higher energy costs may in turn be passed on in higher prices to consumers, thereby possibly reducing demand for their products, or in higher prices to their downstream industries, which could reduce employment demand.

The effects on other energy-intensive sectors are somewhat more complicated to analyse. On the one hand, some of these sectors (basic metals, mineral products like cement, etc.) are an important part of

the supply chains that produce the investment goods. On the other hand, in this scenario the higher carbon price makes the products of these energy-intensive manufacturing sectors more expensive.

In the second decarbonization scenario the amount of additional investment required to achieve the energy-efficiency target is much greater than in the first scenario (€482 billion over 2025–2030 compared to €266 billion). This is reflected in the employment results (see [Table 6](#)).

In 2030, 1,246,000 more jobs are expected in the second scenario than in the reference case. Many of the sectoral patterns are the same. The sectors benefitting directly from the energy-efficiency investment, such as construction and engineering, as well as their supply chains, see large increases in employment. By 2030 the construction sector is expected to have 291,000 more jobs than in the reference case, while engineering and transport equipment together have 152,000 more jobs. Employment in the utilities sector again increases slightly but declines in the extraction sectors.⁷ As in the first decarbonization scenario, the impact on employment in other energy-intensive sectors reflects the net effect of upward and downward pressures. Higher costs lead to decreases in employment, but those sectors that are part of the supply chains that produce the investment goods may increase their employment.

A further key feature in both scenarios is that job reallocations (i.e. job creation and destruction or ‘churn’ between sectors) are projected to take place in other sectors as well as ‘green’ or ‘brown’ sectors. Employment in downstream retail, transport, communications, and business services can be expected to expand more quickly than it would otherwise have done, because changes in their costs and prices induce changes in the composition of final economic demand and hence of labour demand.

Slack in the labour market, or the availability of spare capacity, is observable in the model through the rate of unemployment. In the first decarbonization scenario, EU28 unemployment is 418,000 (1.4%) lower in 2030 than in the reference case, while in the second decarbonization scenario unemployment is 712,000 (3.4%) lower than in the reference case. This decline in unemployment across Europe implies a tightening of the labour market as spare capacity is absorbed. The additional jobs that do not result in lower levels of unemployment come from an expansion of the labour force.

These results for unemployment are based on the implicit assumptions that no skills mismatches exist and that workers can move between the sectors negatively affected by the climate change policies and those that benefit (e.g. ‘green jobs’). In practice, such a change could be associated with changes in skills requirements, which may take time to satisfy. Dedicated government intervention could be needed in order to avoid higher levels of (frictional) unemployment and to limit specific skill shortages. This issue is discussed qualitatively in the context of macroeconomic modelling in Cambridge Econometrics, GHK, IER ([2011](#)).

4.5. The impact of the chosen method of revenue recycling

Previous studies have shown that the method chosen to recycle the revenues is important (see, e.g., Barker, De Ramon, & Pollitt, [2009b](#)). As discussed in Section 3.5 we use a ‘lump-sum’ approach to preserve revenue neutrality. In the first decarbonization scenario each household receives a fixed amount from the carbon tax and ETS revenues, after the investment costs have been subtracted, whereas in the second decarbonization scenario each household pays a fixed charge to cover the investment costs, minus the carbon tax and ETS revenues.

Table 7 EU28 employment (in 1000s of persons, headcount), revenue recycling by adjusting labour taxes

	2010	2015	2020	2025	2030
Reference case	226,149	231,954	233,809	233,365	231,971
<i>S1b – decarbonization (labour tax)</i>	226,149	231,955	233,840	233,789	232,716
Difference from reference case (S1b)	0	1	31	424	745
% Difference (S1b)	0.0	0.0	0.0	0.2	0.3
S1 – decarbonization	Main S1 decarbonization scenario				
Difference from reference case (S1)	0	1	27	563	678
% Difference (S1)	0.0	0.0	0.0	0.2	0.3
<i>S2b – high energy efficiency (labour tax)</i>	226,149	231,956	233,837	233,684	232,616
Difference from reference case (S2b)	0	2	28	319	645
% Difference (S2b)	0.0	0.0	0.0	0.1	0.3
S2 – high energy efficiency	Main S2 decarbonization scenario				
Difference from reference case (S2)	0	2	44	535	1,246
% Difference (S2)	0.0	0.0	0.0	0.2	0.5

Source: E3ME.

Provided that the lump-sum payments are treated as changes in wealth, the effects of revenue recycling are modest in macroeconomic terms. A small proportion of the amount that is recycled as a lump payment in the first decarbonization scenario is used to finance current expenditure. In the second decarbonization scenario, the fixed charge that households must pay has a small negative impact on consumption.

If we had chosen a different method of ensuring revenue neutrality, then the impacts on employment could have been different. As a sensitivity test, changes to labour taxes⁸, which directly affect the cost of labour, were adjusted instead. We found that the employment results in the first scenario were more positive, but they were less positive in the second scenario.

It should be stressed that the results for employment are still positive overall, regardless of the recycling assumption, but the method used changes the order of preference between the scenarios (see [Table 7](#)). The reason for this is that in S1 there is a surplus of revenues which, if used to reduce labour taxes, stimulates employment further. In S2, funding for investment in new equipment must be found and, if this comes through higher labour taxes, it will dampen the positive employment results.

In a final sensitivity we assumed that households' response to the wealth tax is an increase in net saving so that there is no direct change in wealth in the scenarios. This was found to have quite a large impact on the results from S2, with the increase in employment halving from 1,246,000 to 601,000. Again, however, it should be stressed that the impacts are positive overall.

In summary, the positive model results are robust to the revenue recycling assumption, but the exercise shows again that when comparing policy options using a macroeconomic model, the method of revenue recycling is important.

5. Conclusions

This article aimed to provide a macroeconomic assessment of the potential effects on employment of the 2030 climate and energy package compared to a reference case based on current and planned policies to 2020. We used detailed information, drawn from the PRIMES energy model, about the expected investments in the power sector and in energy efficiency improvements as inputs to the macroeconomic E3ME model.

The article has assessed two possible options for the period 2020–2030. In the first, the EU reduces its GHG emissions by 40% (compared to 1990 levels) in the least-cost way, using a uniform CO₂ price across all sectors. In the second, the 40% reduction is combined with a target that renewable energy should have a 30% share of energy consumption and with stricter energy efficiency measures. The article finds that both options result in an increase in net employment by 2030. The second option increases (net) employment by 1.2 million in 2030, compared to the reference case, while the first option increases (net) employment by 0.7 million. Both the positive outcomes and the higher gains in S2 are contrary to previous findings from CGE models.

In our modelling exercise any revenues generated from ETS allowances and carbon taxes (net of investment costs) are recycled under the form of a lump sum payment to, or a fixed charge imposed on, households. Our sensitivity tests show that recycling the (net) revenues to reduce labour taxes (employers' social security contributions) would have a more positive impact on employment than recycling revenues through a lump sum payment under the first option, but would lower the positive impact on employment under the second option. The changes are large enough to reverse the order of preference between the two options.

Overall we can draw three main conclusions from the exercise:

- Both options for meeting the 40% GHG reduction target in 2030 produce higher gross domestic product and employment than the reference case
- The second option, which includes targets for energy efficiency and renewables, has the potential to produce larger increases in employment
- The assumptions about the method of preserving revenue neutrality by recycling additional net revenue or recouping additional net costs can be very important in determining overall results

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Notes

1. In other words, in the modelling framework, levels of economic output affect levels of investment and vice versa. In the version of the model used for this analysis, the econometric parameters suggest that a 1% increase in output could be expected to have a 0.8% increase in investment, while a 1% increase in investment would increase domestic output directly by around 0.08%.
2. This is in part due to the reaction across Europe to the publication of 'Capital in the 21st century' (originally in French, see Piketty, 2013). The wealth tax that has been used in these scenarios is similar in size and nature to the one suggested in the book. An alternative financing method is tested in Section 4.5.
3. The coefficient, given as energy saved per unit of investment, is calculated using energy savings and investment data from the IEA's Efficient World Scenario results for the Organisation for Economic Co-operation and Development (OECD) as a whole. These results are summarized in table 10.3 of IEA (2012, p. 312).
4. See Catte, Girouard, Price, and Andre (2004).
5. Around 80% of the additional investment is in energy efficiency, and the rest is in renewables. It is therefore primarily the energy efficiency that drives the better results in S2.
6. A detailed breakdown by all 69 sectors is also available. For the oil and gas extraction sectors it is assumed that production that is not consumed domestically is exported on world markets, meaning that domestic output and employment levels do not change.
7. The decrease in employment in coal extraction is smaller than in the first decarbonization scenario. This result directly reflects the assumption taken from PRIMES, where coal use in power generation is higher in the second decarbonization scenario than in the first.
8. This is treated as employers' social security contributions. Adjusting employees' social security contributions would be similar to changing income tax rates. In Table 7, the scenarios with the revised assumptions are labelled S1b and S2b.

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Appendix. Labour market equations

This appendix summarizes the structure of the main labour market equations in the E3ME model. Further information can be found in the model manual (Cambridge Econometrics, 2014).

Employment demand

Demand for employment is estimated for 69 sectors in each European Member State. The main long-run factors in determining employment are presented below as average elasticities for the EU. So, for example, a 1% increase in output would lead to an increase in employment of 0.6%.

Employment is a function of

- Economic output (0.6%)
- Real wage rates (– 0.5%)
- Average working hours (– 0.6%)
- Technology, Information and Communication Technology measure (0.0%)
- Technology, other measure (0.0%)

The technology measures are defined as accumulated investment and R&D in each sector; some sectors have a positive impact while others are negative. Further information is available in the model manual.

Labour market participation rates

Labour supply is defined as participation rates multiplied by (exogenous) working age population. Participation rates are estimated for five-year age bands in each EU Member State. Again, the key determinants are presented with average elasticities.

Participation rates are a function of

- Economic output (0.7%)
- Real take-home wage rates (0.6%)
- Unemployment rates (– 0.1%)
- Benefit (or pension) rates (– 0.5%)
- The share of services in output (0.4%)
- Average working hours (– 0.9%)
- The qualifications mix (– 0.3%)

Pension rates are used in the equations for older workers and the share of services in the economy typically affects female participation rates. An indicator for the level of qualifications (high, medium, or low) is included to reflect that better qualified workers are more likely to seek work in some age groups but less likely in others due to the time required to develop qualifications and possible early retirement.